

④ Sub-Array

? Sub Array
 part of an Array → Continuous

→ A single element
 → Entire Array

A: $\begin{matrix} 0 & 1 & 2 & 3 & 4 \\ 3 & 5 & 7 & 2 & 6 \end{matrix}$

idx: [0, 1, 3] → X

idx: [2, 3] → ✓

idx: [2] → ✓

A[2, 4, 1, 6, -3, 7, 8, 9]

1, 6, 8 X

1, 4 X

6, 1, 4, 2 X

7, 8, 9 ✓

A[4, 5, 1, 9, 0, 2, 3, 5]

5 ✓ [acd]
 4, 5, 1, 0 X

0, 0, 2, 3 ✓

4, 5, 1 ✓

→ Unique identifier of a S.A.?

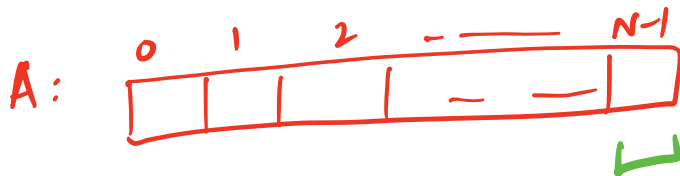
A: $\begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 \\ 4 & 5 & 6 & 2 & 3 & 8 \end{matrix}$

Diagram showing subarray [1, 3] with L=1, R=3.

[L, R] → start & end index of the SA : A[L...R]

$0 \leq L \leq R < N$

Q Given an array of size N .
Find the # of S.A.s



0^{th} idn	1^{st} idn	2^{nd}		$(N-1)^{th}$ idn
$[0, 0]$	$[1, 1]$	$[2, 2]$		$[N-1, N-1]$
$[0, 1]$	$[1, 2]$	$[2, 3]$		
$[0, 2]$	$[1, 3]$			
$\vdots$	$\vdots$			
$[0, N-1]$	$[1, N-1]$	$[2, N-1]$		
$N + N-1 + N-2 + \dots + 1$				

Total # of SA = $N + N-1 + N-2 + \dots + 1$

Total # SA = $\frac{N(N+1)}{2} \approx O(N^2)$

Q Given an Array. Given $[L, R]$ of a S.A.
Print the S.A.

A:

	0	1	2	3	4	5
		^L				^R
A:	5	6	2	5	2	9

print $\rightarrow$ 6, 2, 5, 2, 9

L = 1

R = 5

```
f(i = L; i <= R; i++) {  
    print(A[i]);  
}
```

elements in $[L, R]$
 $= R - L + 1$

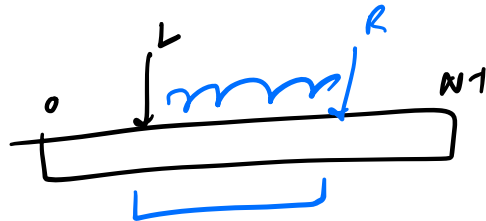
TC = $O(R - L)$

SC = $O(1)$

Q Given an Array. Print all the SA.

$A: [4, 5, 2]$

$[L, R]$	#SA
$[0, 0]$	4
$[0, 1]$	4, 5
$[0, 2]$	4, 5, 2
$[1, 1]$	5
$[1, 2]$	5, 2
$[2, 2]$	2



$$0 \leq L \leq R < N$$

$$N^2 \times N$$

$$TC = O(N^3)$$

```

{ (L = 0; L < N; L++) {
  f(R = L; R < N; R++) {
    // [L, R]
    f(i = L; i <= R; i++) {
      print(A[i]);
    }
    print(new line);
  }
}

```

Given an Array.
Print the sum of All sub Arrays!

A: [4, 5, 2]

[L, R]	# SA	sum
[0, 0]	4	4
[0, 1]	4, 5	9
[0, 2]	4, 5, 2	11
[1, 1]	5	5
[1, 2]	5, 2	7
[2, 2]	2	2

print

I) DP

```

f(L = 0; L < N; L++) {
  f(R = L; R < N; R++) {
    sum = 0

```

```

    f(i = L; i <= R; i++) {
      sum += A[i];

```

```

    }
    print(sum);
  }
}

```

TC = $O(N^3)$

SC = $O(1)$

II) Prefix Sum →

$$\text{sum}(L, R) = \text{ps}[R] - \text{ps}[L-1]$$

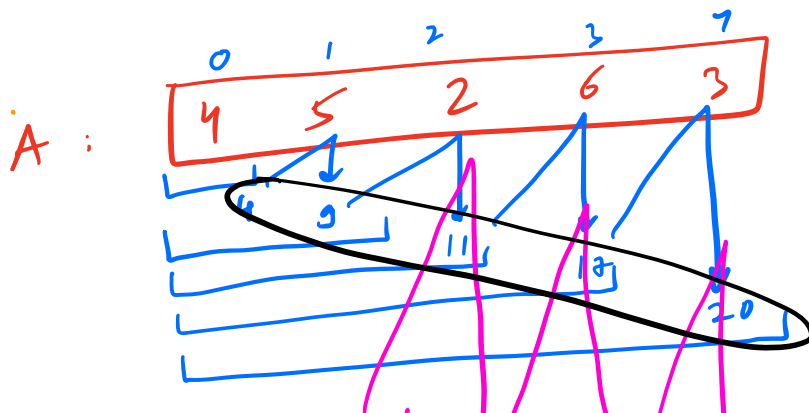
1. Generate $\text{ps}[] \rightarrow O(N)$

2. $\{ \begin{array}{l} f(L=0; L < N; L++) \\ f(R=L; R < N; R++) \end{array} \}$ $\rightarrow N^2 \times O(1)$
 // $[L, R]$
 print($\text{ps}[R] - \text{ps}[L-1]$);
 } $\rightarrow$ $\text{if}(L == 0)$
 }

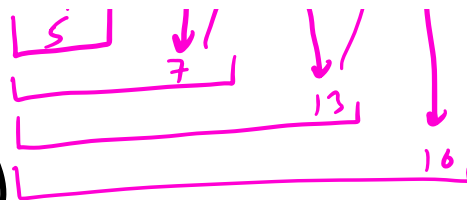
$$\text{TC} = O(N^2)$$

$$\text{SC} = O(N) \rightarrow O(1)$$

III) CARRY FORWARD →



sum = 4
~~4~~
~~8~~
~~11~~
~~15~~
 20



```
f(L = 0; L < N; L++) {
  sum = 0;
```

```
  f(R = L; R < N; R++) {
    sum += A[R];
    print(sum);
```

```
  }
```

```
}
```

N^2
 $O(1)$
 $TC = O(N^2)$
 $SC = O(1)$

Q Given an array. Find the MAX Sub-Array Sum!

ANS = -∞

A: 3 -7 4 2 -5 7

8

```
f(L = 0; L < N; L++) {
  sum = 0;
```

```
  f(R = L; R < N; R++) {
    sum += A[R];
```

```
    ANS = MAX(ANS, sum);
```

```
  }
```

```
  } → return ANS;
```

$TC = O(N^2)$
 $SC = O(1)$

KADANE'S ALGO → TC: $O(N)$
 ↳ MAX SA SUM → SC: $O(1)$

Q Given an Array.
 Find the sum of sum of all subArray!

A: $[1, 3, 5]$

ANS = 0

$[1] \rightarrow 1$
 $[1, 3] \rightarrow 4$
 $[1, 3, 5] \rightarrow 9$
 $[3] \rightarrow 3$
 $[3, 5] \rightarrow 8$
 $[5] \rightarrow 5$
30

```

f(L = 0; L < N; L++) {
    sum = 0;
    f(R = L; R < N; R++) {
        sum += A[R];
        ANS += sum;
    }
}
return ANS;
  
```

TC: $O(N^2)$

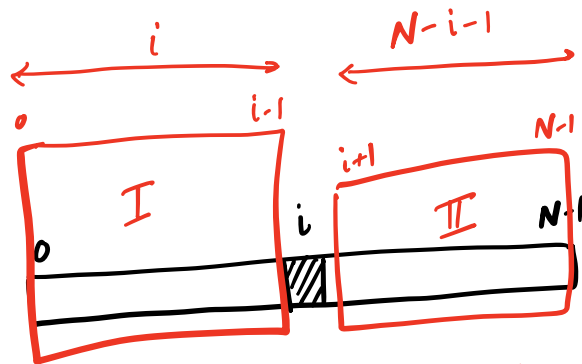
SC: $O(1)$

⑥ CONTRIBUTION TECHNIQUE

$A = [1, 3, 5]$

$(1) \rightarrow 1$
 $+ 4$
 $(1, 3) \rightarrow 4$
 $+ 9$
 $(1, 3, 5) \rightarrow 9$
 $+ 3$
 $(3) \rightarrow 3$
 $+ 8$
 $(3, 5) \rightarrow 8$
 $+ 5$
 $(5) \rightarrow 5$
30

$A[i]$		$(i+1)(N-i)$	
Element	#SAs		Contribution
1	x 3		→ 3
3	x 4		→ 12
5	x 3		→ 15
			<u>30</u>



I)

$$\text{total \#SA} \rightarrow \frac{N(N+1)}{2}$$

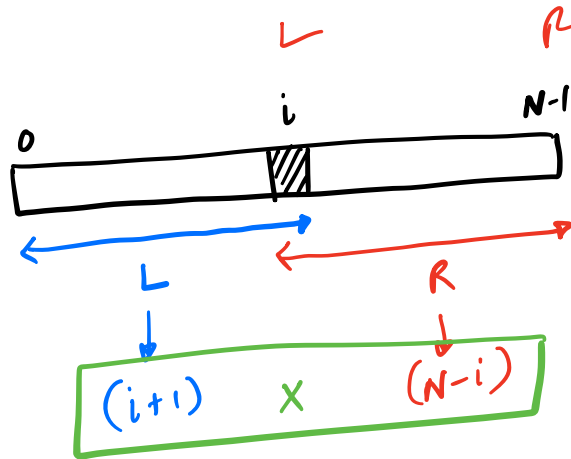
$$\text{\#SAs in I} \rightarrow \frac{i(i+1)}{2}$$

$$\text{II} \rightarrow \frac{(N-i-1)(N-i-1+1)}{2}$$

#SAs which contain the i^{th} element

$$= \frac{N(N+1)}{2} - \frac{i(i+1)}{2} - \left(\frac{(N-i)(N-i-1)}{2} \right)$$

II)



i^{th} element is a part of $(i+1) \times (N-i)$ SAs

$A: [1, 3, 5]$

$N=3$

$i=0 \rightarrow (0+1) \times (3-0) = 3$
 $i=1 \rightarrow (1+1) \times (3-1) = 4$
 $i=2 \rightarrow (2+1) \times (3-2) = 3$

#SAs	Contribution
3	$\times 1$
4	$\times 3 + 3$
3	$\times 5 + 12$
	15
	30

Contribution by i^{th} element $= (i+1) \times (N-i) \times A[i]$

$$ANS = \sum_{i=0}^{N-1} (i+1)(N-i)A[i]$$

```
ANS = 0;  
for (i = 0 → N-1) {  
    ANS += (i+1) × (N-i) × A[i];  
}
```

$N \times O(1)$

return ANS;

TC = $O(N)$
SC = $O(1)$